

SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL ASSIGNMENT TEMPLATE

1. Student Details

Student Name	SEJAL DIPAK TAPASE
PRN	202501110008
Division	AL 1
Batch	A1
Roll No.	202501110008
Date of Submission	4 September 2026

2. Practical Details

Practical No.	1
Practical Title	Class and Objects
Date Performed	13 August 2026
Faculty Name	Khushal Khairnar

3. Problem Statement

Develop a C++ program to create a **Student** class that accepts the **student's roll number, name, and marks** as input and displays the entered details. The program should demonstrate the use of a **class, object, constructor, destructor, and member functions**.

4. Objectives

1. To understand the basic concept of **classes and objects** in C++.
2. To learn how to use a **constructor and destructor**.
3. To create and use **member functions** inside a class.
4. To accept and display **student information** using a C++ program.
5. To understand the basic use of **cin and cout** for input and output.

5. Expected Learning Outcomes:

1. Understand the basic concepts of **classes and objects** in C++.
2. Create a class and define its **data members and member functions**.
3. Understand the working of **constructors and destructors**.
4. Create an object and use it to call **member functions**.
5. Perform basic **input and output operations** using **cin** and **cout**.

6. Theory / Concept

1. Data Members

Data members are variables declared inside a class. They store the information related to an object. In this program, **rollNo**, **name**, and **marks** are data members of the **Student** class.

2. Access Specifier

An access specifier controls where the members of a class can be accessed.

The program uses **public** so that the required member functions can be called using the object.

3. Constructor

A constructor is a special function that is automatically executed when an object is created.

The constructor in this program displays a message when the **Student** object is created.

4. Member Functions

Member functions are functions defined inside a class to perform operations on its data.

The program uses **inputData()** to accept information and **displayData()** to show it.

5. Destructor

A destructor is a special function that is automatically called when an object is destroyed.

The destructor in this program displays a message before the program ends.

6. Object

An object is an instance of a class. It allows us to access the class's data and functions.

Here, **s1** is the object created from the **Student** class.

7. Encapsulation

Encapsulation means keeping related data and functions together inside a class.

The **Student** class combines student information with the functions that operate on it.

This version is **fundamentally different in wording and organization** while still covering the same concepts required by the practical.

7. Algorithm / Pseudocode

Step 1: Start the program.

Step 2: Define a **Student** class with data members for roll number, name, and marks.

Step 3: Create a constructor that displays a message when the object is created.

Step 4: Define **inputData()** to accept the student's roll number, name, and marks.

Step 5: Define **displayData()** to display the stored student information.

Step 6: Define a destructor to display a message when the object is destroyed.

Step 7: Create an object **s1** of the **Student** class.

Step 8: Call **inputData()** using **s1**.

Step 9: Call **displayData()** using **s1**.

Step 10: End the program.

08. Technology / Tools / Libraries Used

Technology / Tool	Purpose
Programming Language	C++
IDE / Platform	DEV CPP
Libraries / Packages	<iostream <string>

09. Implementation / GitHub Repository

Particular	Details
GitHub Repository URL	https://github.com/SejalTapase-AI/CPP_Practical_Assignment_1
Repository Name	CPP_Practical_Assignment_1
Branch Used	main
Date of Last Commit	4 September 2026

10. Test Cases / Input and Output

Test Case	Input	Expected Output	Actual Output	Status
1.	Roll No.: 8 Name: Sejal Marks: 95	Student details should be displayed with the entered values.	Roll No.: 8 Name: Sejal Marks: 95	Pass
2.	Roll No.: 15 Name: Asha Marks: 78	Student details should be displayed with the entered values.	Roll No.: 15 Name: Asha Marks: 78	Pass
3.	Roll No.: 24 Name: Riya Marks: 88	Student details should be displayed with the entered values.	Roll No.: 24 Name: Riya Marks: 88	Pass

11. Results / Observations

The program was successfully executed for all three test cases. The student roll number, name, and marks were accepted as input and displayed correctly. The constructor message appeared when the object was created, and the destructor message appeared when the object was destroyed.

Result: The program successfully demonstrates the use of **class, object, constructor, destructor, and member functions** in C++.

```
Constructor called.  
Enter Roll No.: 8  
Enter Name: Sejal  
Enter Marks: 95  
  
Student Details:  
Roll No.: 8  
Name: Sejal  
Marks: 95  
Destructor called.  
  
-----  
Process exited after 17.66 seconds with return value 0  
Press any key to continue . . . |
```

```
Constructor called.  
Enter Roll No.: 15  
Enter Name: Asha  
Enter Marks: 78
```

```
Student Details:  
Roll No.: 15  
Name: Asha  
Marks: 78  
Destructor called.
```

```
-----  
Process exited after 42.14 seconds with return value 0  
Press any key to continue . . . |
```

12. Conclusion

This practical helped in understanding the basic concepts of **Object-Oriented Programming in C++**. A **Student** class and its object were used to store and display student details. The practical also demonstrated how **constructors, destructors, and member functions** work during program execution.

13. Assignment Questions

Q1. What is a class?

Answer:

A class is a blueprint used to create objects. It contains data members and functions.

Q2. What is an object?

Answer:

An object is an instance of a class. It is used to access the members of the class.

Q3. What is the main purpose of a constructor?

Answer:

A constructor is used to perform initialization when an object is created.

Q4. When does a destructor execute?

Answer:

A destructor executes automatically when an object is destroyed or goes out of scope.

Q5. What are member functions?

Answer:

Member functions are functions that belong to a class and operate on its data.

Q6. What does **cin** do?

Answer:

cin is used to take input from the user.

Q7. What does **cout** do?

Answer:

cout is used to display output on the screen.

14. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name	PRN	Signature
SEJAL DIPAK TAPASE	202501110008	